

Some helpful definitions...

Working Directory

- After installing R & RStudio, you need to set the working directory
 - This is the location on your computer where any data files to imported can be found, and where any R scripts (the files that save your code) will be saved
- In R studio, you can set the working directory with the menus (*Session >> Set Working Directory >> Choose Directory*) or with a line of code that gives the path of the folder on your computer:
 - `>setwd("Drive:/Folder1/Folder2")`
- If you have made a new R Project from an existing Directory as shown in the live demo, working directory will be set this way  Projects are always recommended! 
- can always check working directory with the function:
 - `>getwd()`

Some vocabulary

- **Function** = how you get stuff done in R
- **Argument** = specifications of functions
- **Packages** = are a collection of R functions, complied code and sample data. By default a set of packages are installed during installation. They are stored under a directory called "library" in the R environment
- **Documentation** = the explanations of functions and arguments for different packages written by the authors
- **“run” or “running code”** = enter command into the R console to make it happen
- **Script** = a text file containing (almost) the same commands that you would enter on the command line of R
- **Data frame** = a *special type of list* where every element of the list has same length (i.e. data frame is a “rectangular” list); *de facto* data structure for most tabular data and what we use for statistics.
 - **Tibble** = tidyverse style dataframe
- **Indexing** = selecting a subset of the elements in order to use them in further analysis or possibly change them. Style depends on syntax

Variable types in R

- **character**: " a ", " swc "
- **numeric**: 2, 15 . 5
 - “continuous” variable in other software
- **integer** is similar, use numeric in practice (less limitations)
- **double** is similar, but is a numeric value with decimal places
- **logical**: TRUE, FALSE
- **complex**: $1+4i$ (complex numbers with real and imaginary parts)
- **factor**: used to describe items that can have a finite number of values (gender, social class, etc.).
 - A factor has a `levels` attribute and class "factor"
 - Optionally, it may also contain a `contrasts` attribute which controls the parametrization used when the factor is used in a modeling functions.
 - “categorical” variable in other software. Tell R that a variable is **nominal** by making it a factor. An ordered factor is used to represent an **ordinal variable**.

Data structures in R

- **Vectors** = most common and basic structure in R; a collection of elements that are most commonly of mode character, logical, integer or numeric.
- **Atomic vector** = vector where elements must be the same data type; default vector type
- **List** = a special type of vector. Each element can be a different type.
- **Matrix** = an extension of the numeric or character vectors. They are not a separate type of object but simply an atomic vector with dimensions; the number of rows and columns. As with atomic vectors, the elements of a matrix must be of the same data type.
- **Array** = similar to matrices but can have more than two dimensions
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